

Puneet Velidi

Statistical AI/ML Researcher · Founder, ChimeraML

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EDUCATION

University of Victoria

M.Sc. in Statistics (thesis-based, fully funded)

Co-supervisors: Prof. Farouk Nathoo & Prof. Michelle Miranda. Thesis: *Methods for Censored Covariates, Spatial Transcriptomic Data and Neural Manifolds*.

Victoria, BC

Jan. 2025 – Dec. 2026

Cornell University

B.A. in Statistics & Economics

Ithaca, NY

Sep. 2018 – Dec. 2022

PUBLICATIONS († joint first authorship)

P. Velidi, M. Miranda, F. Nathoo, A. Mbilinyi, and C. Beaulac. *Shape Analysis as an Inductive Bias for Glioma Grading from Tumor Contours*. Submitted for publication. 2026. [arXiv](#)

P. Velidi, E. Amico, F. Nathoo, and M. Miranda. *Neural manifold connectomics reveals multiregime functional connectivity*. **Under review**. 2026. [bioRxiv](#)

P. Velidi, M. Griffin, J. Liu, et al., F. Nathoo. *SPARTACCUS: Semiparametric Bayesian Proportional Hazards Model with a Censored Covariate in Huntington's Disease*. In preparation. 2026.

P. Velidi, Z. Wei, and F. Nathoo. *Covariance Nonstationarity is Evident in Spatial Transcriptomics and Provides a New Categorization of Spatially Varying Genes*. Submitted for publication. 2026. [bioRxiv](#)

F. Luchese, **P. Velidi**, et al., P. Cassano. *Transcranial Photobiomodulation on Language and Cognitive Performance in Down Syndrome: A Pilot Randomized Sham-Controlled Trial*. **Under review**. 2026. [medRxiv](#)

B. Ferreras ... **P. Velidi**, et al., P. Cassano. *Transcranial Photobiomodulation in Bipolar Disorder: A Case Series on Executive Skills and fMRI BOLD Signal*. Submitted for publication. 2026. [SSRN](#)

G. Gonzalez-Garibay, **P. Velidi**, et al., P. Cassano. *Volumetric Changes Induced by Transcranial Photobiomodulation Therapy in Major Depressive Disorder: An Exploratory Neuroimaging Study*. Submitted for publication, 2026. [SSRN preprint](#)

P. Velidi[†], Z. Wei[†], et al., F. S. Nathoo. *Generalized Jeffreys's Approximate Objective Bayes Factor: Model Selection Consistency, Finite-Sample Accuracy, and the Bayesian Evidence in 71,126 Clinical Trial Findings*. **Under revision**. 2025. [arXiv](#).

S. P. Singleton[†], **P. Velidi**[†], et al., A. Kuceyeski. *Altered structural connectivity and functional brain dynamics in individuals with heavy alcohol use elucidated via network control theory*. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, 2024.

ACADEMIC AWARDS & SCHOLARSHIPS

Labs4 Technology Readiness Level-Up Fellowship — 15,000 CAD (2025).

UVic Betty & Gilbert Kennedy Graduate Scholarship in Mathematics & Statistics — 5,130 CAD (2025-2026).

Maud Menten Institute / PIMS Research Accelerator Award — 5,000 CAD (2025).

UVic FGS Graduate Fellowship, Master's — 5,000 CAD (2025-2026).

SELECTED INDUSTRY EXPERIENCE

Founder — ChimeraML

Deeptech research studio

Vancouver, BC & Washington, DC

Sept. 2025 – Present

- Backed by NVIDIA Inception, Labs4, and Mitacs.

Software Engineer II — Walmart Global Tech

Enterprise Business Services – Associate Digital Experience

Bentonville, AR

Feb. 2023 – May 2024

- Early Career Recognition, awarded to <10% of the 2023 Walmart Global Tech Development Program.

RESEARCH EXPERIENCE

Graduate Researcher — University of Victoria, Dept. of Mathematics & Statistics

Nathoo Lab, PIs: Prof. Farouk Nathoo & Prof. Michelle Miranda

Victoria, BC

Jan. 2025 – Present

Visiting Researcher — University of Birmingham, School of Mathematics

AMICO Lab, PI: Prof. Enrico Amico

Birmingham, UK

Aug. 2025

Postbacc Researcher — Harvard Medical School & MGH, Neuropsychiatry & Neuromodulation

MGH Photobiomodulation Laboratory, PI: Prof. Paolo Cassano

Boston, MA

May 2024 – Dec. 2024

- Statistical analysis of the TransPhoM-DS, TRIADE, TPEB, and TRAP-AD clinical trials.

Undergraduate Researcher — Weill Cornell Medicine, Dept. of Radiology

Computational Connectomics Laboratory, PI: Prof. Amy Kuceyeski

Ithaca, NY

Jan. 2022 – Dec. 2022

SELECTED INVITED TALKS & SESSIONS

Invited Session Chair & speaker, 7th ICSA-Canada Chapter Symposium (2026) — Neural manifold connectomics reveals multiregime functional connectivity.

Invited Seminars: “Applications of Protein Folding Models in Immunotherapeutic Research,” UBC Math Bio (2026); “70,000 Leagues Under the Sea of p -Values,” UBC Statistics & Maud Menten Institute (2025); “Network Control Theory in Neuroscience,” UVic Cognition & Brain Sciences and Maud Menten Institute (2025).

TECHNICAL REPORT

R. Adenane, J. Beaudin, L. Kankanamge, and **P. Velidi**. *Immunodynamic Modeling in Melanoma and Gastric Cancer*. 2025.

POSTER PRESENTATIONS (full titles and posters)

Photobiomodulation and glymphatic CSF dynamics in MCI / early Alzheimer disease — Clinical Research Day, 2026 (upcoming).

Bayesian survival analysis with a right-censored covariate — ICSA-Canada, 2026.

Bayesian analysis of the TransPhoM-DS clinical trial — OHBM, 2026.

Ahab's Harpoon: calibrated irreproducibility detection with approximate objective Bayes factors — SSC, 2026.

Reassessing clinical-trial evidence with extended approximate objective Bayes factors — MMI HQP Summit, 2025.

Photobiomodulation-induced volumetric brain changes in depression — CNS, 2025.

Structural connectivity and brain dynamics in heavy alcohol use via network control theory — OHBM, 2024.

Network control analysis of individuals with substance use disorder — OHBM, 2023.